

Sam Bright-Thonney

IAIFI Fellow, Massachusetts Institute of Technology

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APPOINTMENTS

Massachusetts Institute of Technology

2024 - Present

- Postdoctoral Fellow, NSF AI Institute for Artificial Intelligence and Fundamental Interactions ([IAIFI](#))
- Fellowship appointment: September 2024 - August 2027

EDUCATION

Cornell University

2018 - 2024

- Ph.D. in Experimental Particle Physics. Defended June 26th, 2024.
- Committee: Julia Thom-Levy (Advisor), Jim Alexander, Maxim Perelstein
- [Thesis](#): *Dark Photons & Deep Learning: Unconventional Searches for New Physics at the Large Hadron Collider*

University of California, Berkeley

2014 - 2018

- B.A., Physics. B.A., Applied Mathematics
- *Magna Cum Laude*, Phi Beta Kappa

PUBLICATIONS & PREPRINTS

1. S. Bright-Thonney, T. R. Harvey, A. Lukas, and J. Thaler, *Sven: Singular Value Descent as a Computationally Efficient Natural Gradient Method*, [arXiv:2604.01279](#)
2. E. A. Moreno, S. Bright-Thonney, A. Novak, D. Garcia, and P. Harris, *AI Agents Can Already Autonomously Perform Experimental High Energy Physics*, [arXiv:2603.20179](#)
3. S. Bright-Thonney, C. Reissel, G. Grosso, N. Woodward, K. Govorkova, A. Novak, S. E. Park, E. Moreno, and P. Harris, *AutoSciDACT: Automated Scientific Discovery through Contrastive Embedding and Hypothesis Testing*, **NeurIPS 2025 Poster** [[arXiv:2510.21935](#)]
4. S. Bright-Thonney, *Embedding Jets with Maximum Manifold Capacity Representations*. **Machine Learning for the Physical Sciences Workshop, NeurIPS 2025**. [[OpenReview](#)]
5. **CMS** Collaboration, *Machine-learning techniques for model-independent searches in dijet final states*. (2025), [[arXiv:2512.20395](#)]
6. **CMS** Collaboration, *Model-agnostic search for dijet resonances with anomalous jet substructure in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *Reports on Progress in Physics* **88** (June, 2025) 067802
7. S. Bright-Thonney, B. Nachman, and J. Thaler, *Infrared-safe energy weighting does not guarantee small nonperturbative effects*, *Phys. Rev. D* **110** (2024), no. 1 014029, [[arXiv:2311.07652](#)] **Editor's Selection**
8. S. Bright-Thonney, P. Harris, P. McCormack, and S. Rothman, *Chained quantile morphing with normalizing flows*, [arXiv:2309.15912](#)
9. **CMS** Collaboration, *Search for inelastic dark matter in events with two displaced muons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *Phys. Rev. Lett.* **132** (Jan, 2024) 041802
10. S. Bright-Thonney, I. Moutl, B. Nachman, and S. Prestel, *Systematic quark/gluon identification with ratios of likelihoods*, *JHEP* **12** (2022) 021, [[arXiv:2207.12411](#)]

11. S. Bright-Thonney and B. Nachman, *Investigating the Topology Dependence of Quark and Gluon Jets*, *JHEP* **03** (2019) 098, [[arXiv:1810.05653](#)]
12. **ATLAS** Collaboration, M. Aaboud et al., *Search for R-parity-violating supersymmetric particles in multi-jet final states produced in p-p collisions at $\sqrt{s} = 13$ TeV using the ATLAS detector at the LHC*, *Phys. Lett. B* **785** (2018) 136–158, [[arXiv:1804.03568](#)]

CONFERENCES, POSTERS, & INVITED TALKS

♠ = Invited Talk, ■ = Talk, □ = Poster

- “[AutoSciDACT: Automated Scientific Discovery through Contrastive Embedding and Hypothesis Testing](#)”. *NeurIPS*. San Diego, CA. December 2025
- “[Embedding Jets with Maximum Manifold Capacity Representations](#)”. *Machine Learning and the Physical Sciences Workshop, NeurIPS*. San Diego, CA. December 2025
- “[Supervised contrastive learning for streamlined analysis & anomaly detection](#)”. *7th Inter-Experimental LHC Machine Learning Workshop*. CERN (Geneva, CH). May 2025.
- ♠ “[Collide and compress: building robust embedding spaces to simplify new physics searches at the LHC](#)”. *IAIFI Colloquium*. MIT, February 2025. ([Slides](#))
- ♠ “[Towards Robust AI in HEP](#)”. *Physics in the AI Era Workshop*. Pisa, Italy. September 2024.
- ♠ “[Adventures in the LHC’s ‘Wild West’ Era](#)”. *Dicke Fellowship Finalist Seminar*. Princeton, November 2023.
- ♠ “[Adventures in the LHC’s ‘Wild West’ Era](#)”. *Lederman Fellowship Finalist Seminar*. Fermilab, November 2023.
- ♠ “[Wrangling the Black Box](#)”. *UC Irvine Particle Physics/ML Seminar*. Virtual, August 2023.
- “[Chained Quantile Morphing with Normalizing Flows](#)”. *BOOST 2023*. Berkeley, CA, August 2023.
- “[Search for inelastic dark matter with the CMS detector](#)”. *Phenomenology 2023 Symposium*. Pittsburgh, PA, May 2023.
- “[Design and Construction of Radiation-Hard, Thermally Conductive Dees for the HL-LHC Upgrade to the CMS Inner Tracker](#)”. *APS April Meeting*. New York, NY, April 2022. [Slides](#)
- “[Efficiently Cooling Radiation-Damaged Silicon Pixel Sensors in the CMS HL-LHC Inner Tracker](#)”. *54th Annual Fermilab Users Meeting*. Virtual, August 2021

HONORS & AWARDS

2023	Finalist, IAIFI Fellowship (Awarded)	<i>IAIFI (MIT)</i>
2023	Finalist, Pappalardo Fellowship	<i>MIT, Physics Department</i>
2023	Finalist, Dicke Fellowship	<i>Princeton, Physics Department</i>
2023	Finalist, Lederman Fellowship	<i>Fermilab</i>
2022, 2023	Albert Silverman Memorial Travel Award	<i>Cornell University, Physics Dept.</i>
2022	FGSA Travel Award, April Meeting	<i>APS</i>
2021	Fermilab Guests & Visitors Program	<i>Fermilab</i>
2018	Hartman Fund Teaching Scholarship	<i>Cornell University, Physics Dept</i>
2017-18	Berkeley Physics Undergraduate Research Scholars	<i>UC Berkeley, Physics Dept</i>
2017	URAP Chandra Research Fellow	<i>UC Berkeley</i>

TEACHING EXPERIENCE

August 2025	IAIFI Summer School Tutorial: <i>Computing with Neural Manifolds</i> (Code)	<i>IAIFI (MIT)</i>
Spring 2020	ENGRD 2700 (Intro Statistics)	<i>Cornell University</i>
Fall 2019	Physics 2213 (Intro E&M)	<i>Cornell University</i>
Fall 2018, Spring 2019	Physics 1112 (Intro Mechanics)	<i>Cornell University</i>

OUTREACH, LEADERSHIP & SERVICE

Journal Reviewer 2024 - Present

- Reviews for Physical Review D (PRD) and Physical Review Letters (PRL)

REACT Workshop July 2022, August 2023

- Cornell's Research Education and Activities for Community Teachers workshop. Presented an [introduction to pixel detectors](#) (2022) and made a web-hosted [Jupyter demo](#) on the physics of sound and digital audio (2023).

Particle Bites 2022 - 2024

- Writer for [ParticleBites](#); covered the [CMS Higgs width measurement](#) and the [CDF W mass measurement](#).

Cornell Prison Education Program January - August 2022

- Taught math & tutored writing at Auburn Correctional Facility

Graduate Student Representative, CEDI Committee August 2021 - July 2022

- Cornell Physics department's Committee on Equity, Diversity, and Inclusion (CEDI)

Cornell Physics Admissions Committee January - February 2022

- Graduate student representative on the Cornell Physics PhD admissions committee for Fall 2022

Physics Application Workshop November 2020, 2021

- Lead organizer of two virtual PhD application workshops for the fall 2020 and 2021 application seasons.

Graduate & Professional Student Association September 2020 - May 2023

- Served as a physics field representative (2020-21), elected Physical Sciences Voting Member (2021-23)

Physics Graduate Society May 2019 - August 2020

- Communications Officer for PGS; organized and advertised department social events

TECHNICAL SKILLS

Programming Languages	Python, C++, MATLAB
Software Packages	PyTorch, Jax, ROOT, Scientific Python Libraries
Physics Tools	PYTHIA 8, DIRE, MadGraph5, Herwig, CMSSW
Tools	Git, Unix Shell, Claude Code